PROJECT SYNOPSIS

Facial Recognition Based Employee Monitoring System

by

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1.Title of the Project

Facial Recognition Based Employee Monitoring System

2. Introduction and Background of the Study

Traditional manual attendance monitoring is inefficient in modern workplaces. So, this project addresses that by automating monitoring through AI-powered video analysis, improving productivity oversight and enabling data-driven HR decisions.

Facial Recognition based Employee Monitoring Systems use combination of facial recognition and object detection (YOLOv8) with OpenCV Library for real time employee and people(inside & outside) presence and absence detection in a defined workplace .

3. Objectives of the Project

- Employee registration module that take user details and captures 50 high-quality facial samples per individual using real-time webcam feed.
- Detect employee and people (inside & Outside desk area) presence/absence in a defined desk area with high accuracy.
- To perform facial recognition of employees and people around the desk area using KNN classifier trained on captured face samples.
- To generate detailed logs of entry and exit of people(inside & Outside desk area) and absence durations for employee inside the desk area.

4. Problem Statement / Research Question

How can workplaces accurately and non-intrusively track employee workstation occupancy to measure attendance and detect prolonged absences without manual intervention?

5. Methodology / Approach

Step	Methodology / Approach	Description
1	**Video Capture**	Capture object images through a webcam in real-time for object detection.

Step	Methodology / Approach	Description
2	**Face Detection**	Use algorithms like Haar Cascade detectors to detect faces in the frame.
3	**Attendance marking**	Recording Attendance of employees
4	**Desk Area Detection**	Auto-detects monitoring area using desk objects using Object detection YOLOv8.
5	**Person Detection**	Detects persons with confidence filtering using Object detection YOLOv8**.**
6	**Presence Analytics**	It calculates the "overlap ratio" of a person relative to the desk area to Tracks presence of people(inside or outside) and absence of employee with configurable absence duration.
7	**Data Logging**	To generate automated CSV reports for detection events and precise absence durations

6. Tools, or Technologies to be Used

A. Backend:

1.Python: Most commonly used language for computer vision and machine learning tasks.

2.OpenCV: An open-source computer vision library often used for image processing and face detection tasks.

3.Numpy: Numpy is a fundamental package for numerical computing in Python. Etc.

4.Ultralytics: Ultralytics is a popular open-source Python library that provides YOLO models for computer vision tasks like object detection.

5.yolov8n.pt: It is the nano variant of Ultralytics YOLOv8 detection model which is pre-trained on COCO dataset.

B. Frontend:

1.GUI: A GUI of handling all the task like Employee registration , Attendance recording, Employee monitoring and view Logs and Attendance.

7. Expected Outcomes / Deliverables

- **Outcome**: A fully functional Facial Recognition based Employee Monitoring Systems with real-time presence status, absence duration, people detection, and event logs.

8. Timeline of Work (in phases along with completion dates of each task)

Phase-1

Sr. No	Task	Description	Completion Date
1	Technology Selection	Import necessary libraries and modules: OpenCV (cv2), NumPy, threading, datetime. **YOLOv8** for object detection and **Haar Cascade** for face detection .	**20 Feb 2025**
2	Employee Tracker Class	Initialize Model Loading , Logging System and core class with state variables (presence status, config params like confidence and ROI Setup)	**1 March 2025**
3	Desk Area Detection	Develop a function for Execute multi-frame analysis for desk objects like chairs, tables, and laptops to define the ROI.	**20 March 2025**

Phase-2

4	Human Detection	Implement real-time YOLOv8 inference to isolate the "Person" class within the video feed.	**30 March 2025**
5	Identity Verification	Integrate object detection using Haar Cascades and KNN to match detected faces against the employee database.	**10 April 2025**
6	Spatial Analytics	Develop the overlap logic to calculate if a person is "Inside" or "Outside" the desk boundary based on bounding box coordinates.	**20 April 2025**
7	Data Logging	it creates a daily log folder and CSV files for both general detections and specific absence durations.	**01 May 2025**

9. References (if any)

1. Blogs related to this project (computer vision & Object detection)

2.Documentation regarding different libraries and modules

3. YouTube videos